10.5 Solution of Ordinary Differential Equations

Definition 10.9 : (Solution of DE)

A **solution** of a differential equation is an expression for the dependent variable in terms of the independent variable(s) which satisfies the differential equation.

Caution

(i) There is no guarantee that a differential equation has a solution.
For instance, $\left(y'(x)\right)^2 + y^2 + 1 = 0$ has no solution, since $\left(y'(x)\right)^2 = -\left(y^2+1\right)$ and so $y'(x)$ cannot be real.

(ii) Also, a solution of a differential equation, if exists, is not unique.
For instance, the functions $y = e^{2x}$, $y = 2e^{2x}$, $y = \sqrt{8}e^{2x}$ are solutions of same equation $\frac{dy}{dx} - 2y = 0$. In fact, $y = ce^{2x}$, $c \in \mathbb{R}$, are all solutions of the differential equation $\frac{dy}{dx} - 2y = 0$.
Thus, to represent all possible solutions of a differential equation, we introduce the notion of the general solution of a differential equation.

Definition 10.10 : (General solution)

The solution which contains as many arbitrary constants as the order of the differential equation is called the **general solution**

Remark

The **general solution** includes all possible solutions and typically includes arbitrary constants (in the case of an ODE) or arbitrary functions (in the case of a PDE.)

Definition 10.11 : (Particular solution)

If we give particular values to the arbitrary constants in the general solution of differential equation, the resulting solution is called a Particular Solution.

Remark

(i) Often we find a particular solution to a differential equation by giving extra conditions.

(ii) The general solution of a first order differential equation $y' = f(x, y)$ represents a one-parameter family of curves in xy-plane.
For instance, $y = ce^{2x}$, $c \in \mathbb{R}$, is the general solution of the differential equation $\frac{dy}{dx} - 2y = 0$.
For instance, we have already seen that $y = a\cos x + b\sin x$ satisfies the second order differential equation $\frac{d^2y}{dx^2} + y = 0$. Since it contains two arbitrary constants, it is the general solution of $\frac{d^2y}{dx^2} + y = 0$. If we put $a = 1$, $b = 0$ in the general solution, then we get $y = \cos x$ is a particular solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$.

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